Name: Miheeka Kewalramani Roll No.45 Branch:AI&DS Class:T12 Subject:ASD&D

EXPERIMENT:8

Aim: To implement CI/CD pipeline to host a website on AWS

Lab Outcome: LO3

LO3: Build and manage automated CI/CD pipelines for application development, testing, packaging, and deployment.

Theory:

Continuous Integration and Continuous Deployment (CI/CD)

Continuous Integration (CI) is a software development practice where developers merge their code changes frequently into a central shared repository, followed by automated builds and automated test executions to detect integration bugs early. Continuous Delivery/Deployment (CD) extends this practice by automating the release process so that verified code changes can be automatically packaged and deployed directly to staging or production environments without manual intervention.

Implementing CI/CD drastically shortens the software development lifecycle (SDLC), minimizes manual deployment errors, improves software stability, and enables rapid iteration of software products.

Key Components of AWS CI/CD Architecture

- Source Repository (GitHub): Acts as the centralized version control system (VCS). Webhooks or OAuth connection points alert the cloud pipeline instantaneously whenever commits are pushed to specific branches (e.g., main).
- AWS CodePipeline: A continuous delivery orchestration service that models, visualizes, and coordinates every step of the release process. It manages the stages—

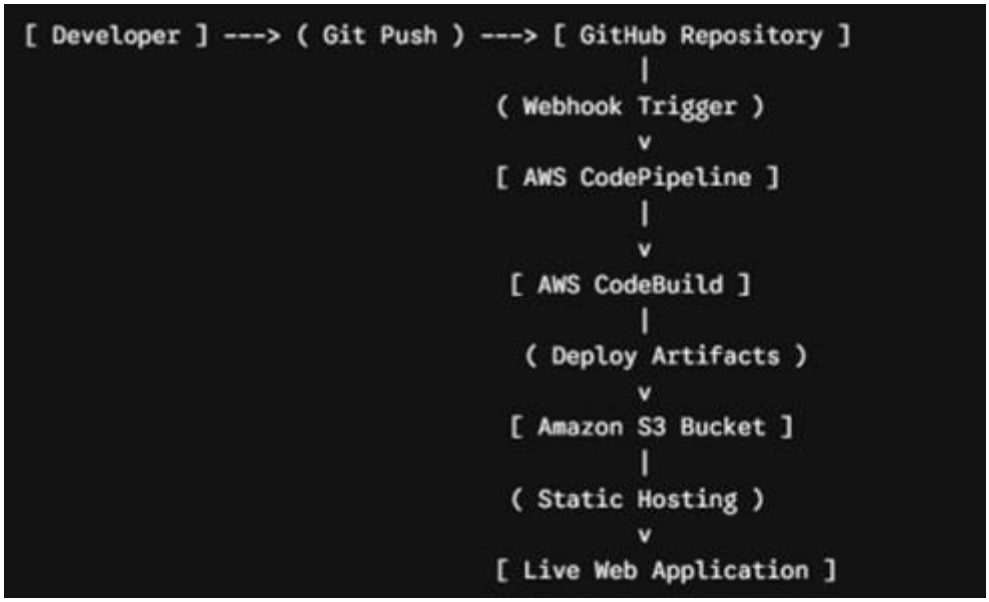
such as Source, Build, and Deploy—moving build artifacts sequentially through execution pipelines.

- AWS CodeBuild: A fully managed, scalable build and compilation service. CodeBuild spins up isolated ephemeral compute containers, runs configuration scripts specified in a buildspec.yml file, runs unit tests, compiles static assets, and bundles output artifacts for downstream deployment.
- Amazon Simple Storage Service (Amazon S3): Provides scalable object storage with built-in Static Website Hosting. When configured with public read access (s3:GetObject) and an index.html root document, it serves as a lightweight, highly durable, serverless web server directly from bucket storage.
- AWS Identity and Access Management (IAM): Enforces fine-grained least-privilege security. Service roles ensure CodeBuild and CodePipeline have the exact permissions required (such as read/write access to S3 artifact and deployment buckets) to operate securely without exposing root or persistent user credentials.

Operational Workflow

- Push: A developer commits changes to the GitHub repository.
- Trigger: GitHub triggers an automated event notification to AWS CodePipeline via webhook integration.
- Build: CodePipeline retrieves the source files and triggers AWS CodeBuild to execute containerized compilation and packaging.
- Deploy: The pipeline extracts the verified build artifacts directly into the designated S3 deployment bucket.
- Serve: The Amazon S3 static website hosting endpoint renders the live web application immediately to users.

Architecture & Workflow:



Implementation:

Phase 1: AWS CodeBuild Project Configuration

Step 1: Create a Build Project

1. Open the AWS Management Console and navigate to Developer Tools > CodeBuild.
2. Click on Create build project.
3. Under Project configuration, set the Project name to pwa-codebuild-vb.
4. Select Default project as the project type.

aws

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Root user email address

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miheeka27@gmail.com

AWS account name

Choose a name for your account. You can change this name in your account settings after you sign up.

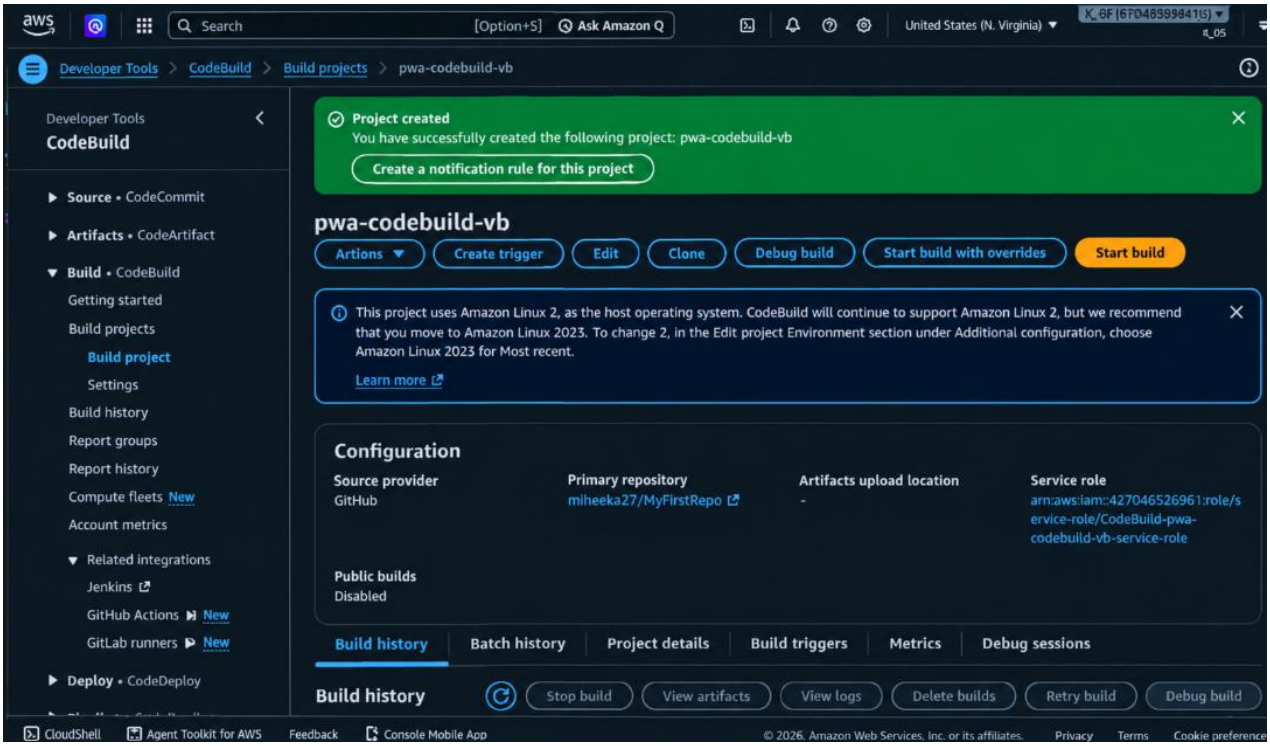
Miheeka Kewalramani

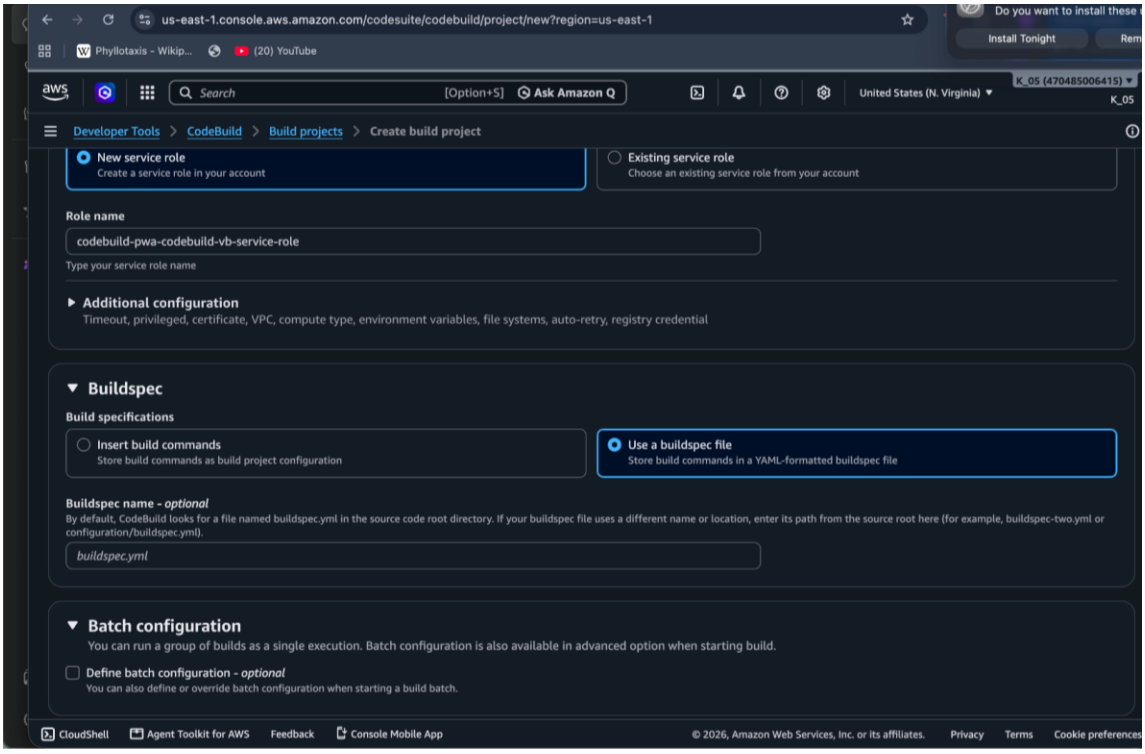
Verify email address

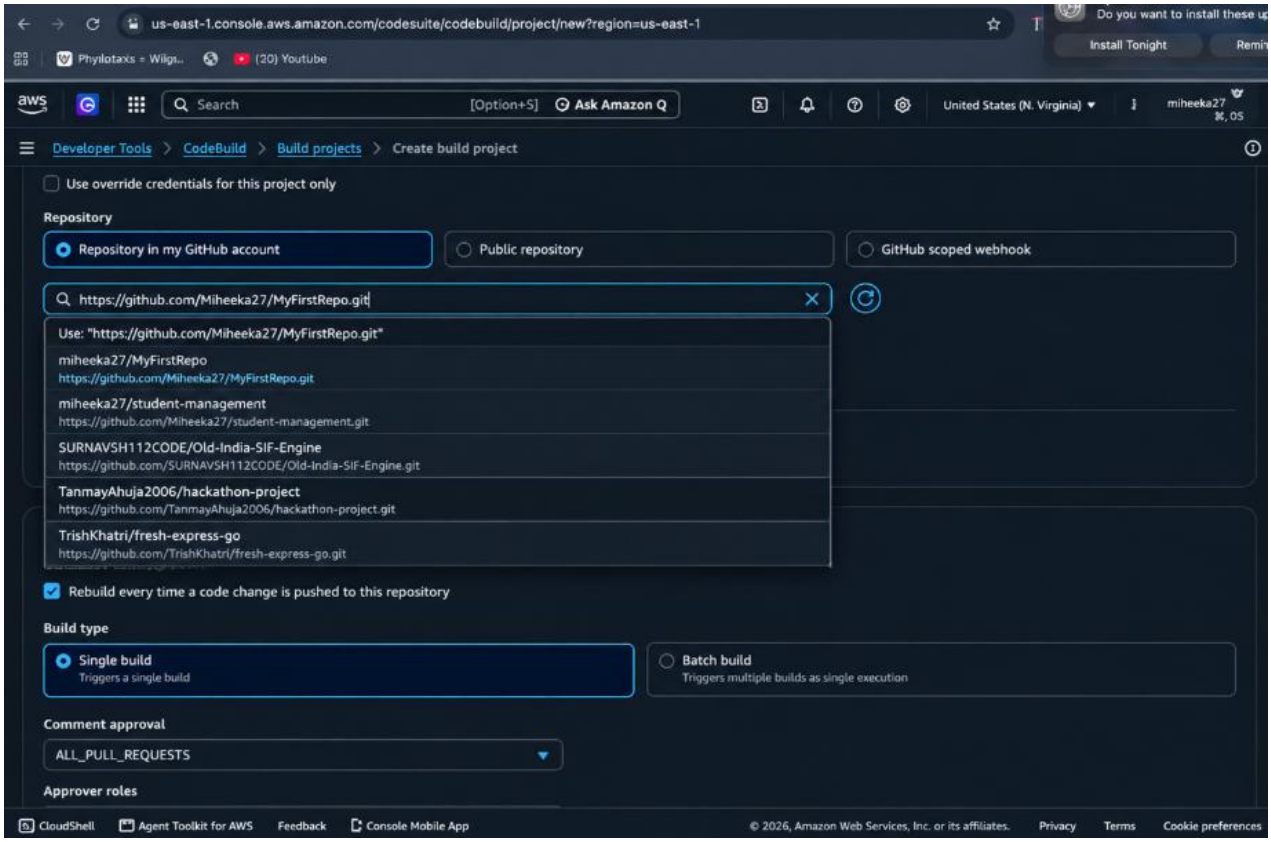
OR

Sign in to an existing AWS account

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Search
[Option+S]
Ask Amazon Q
United States (N. Virginia)
Developer Tools > CodeBuild > Build projects > pwa-codebuild-vb
Developer Tools
CodeBuild
Source • CodeCommit
Artifacts • CodeArtifact
Build • CodeBuild
Getting started
Build projects
Build project
Settings
Build history
Report groups
Report history
Compute fleets New
Account metrics
Related integrations
Jenkins
GitHub Actions New
GitLab runners New
Deploy • CodeDeploy
Project created
You have successfully created the following project: pwa-codebuild-vb
Create a notification rule for this project
pwa-codebuild-vb
Artions
Create trigger
Edit
Clone
Debug build
Start build with overrides
Start build
This project uses Amazon Linux 2, as the host operating system. CodeBuild will continue to support Amazon Linux 2, but we recommend that you move to Amazon Linux 2023. To change 2, in the Edit project Environment section under Additional configuration, choose Amazon Linux 2023 for Most recent.
Learn more
Configuration
Source provider
GitHub
Primary repository
miheeka27/MyFirstRepo
Artifacts upload location
-
Service role
arn:aws:iam::427046526961:role/service-role/CodeBuild-pwa-codebuild-vb-service-role
Public builds
Disabled
Build history
Batch history
Project details
Build triggers
Metrics
Debug sessions
Build history
Stop build
View artifacts
View logs
Delete builds
Retry build
Debug build
CloudShell
Agent Toolkit for AWS
Feedback
Console Mobile App
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us-east-1.console.aws.amazon.com/codesuite/codebuild/project/new?region=us-east-1
Search
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United States (N. Virginia)
Developer Tools > CodeBuild > Build projects > Create build project
New service role
Create a service role in your account
Existing service role
Choose an existing service role from your account
Role name
codebuild-pwa-codebuild-vb-service-role
Type your service role name
Additional configuration
Timeout, privileged, certificate, VPC, compute type, environment variables, file systems, auto-retry, registry credential
Buildspec
Build specifications
Insert build commands
Store build commands as build project configuration
Use a buildspec file
Store build commands in a YAML-formatted buildspec file
Buildspec name - optional
By default, CodeBuild looks for a file named buildspec.yml in the source code root directory. If your buildspec file uses a different name or location, enter its path from the source root here (for example, buildspec-two.yml or configuration/buildspec.yml).
buildspec.yml
Batch configuration
You can run a group of builds as a single execution. Batch configuration is also available in advanced option when starting build.
Define batch configuration - optional
You can also define or override batch configuration when starting a build batch.
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us-east-1.console.aws.amazon.com/codesuite/codebuild/project/new?region=us-east-1
Search
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Ask Amazon Q
United States (N. Virginia)
Developer Tools
CodeBuild
Build projects
Create build project
Use override credentials for this project only
Repository
Repository in my GitHub account
Public repository
GitHub scoped webhook
https://github.com/Miheeka27/MyFirstRepo.git
Use: "https://github.com/Miheeka27/MyFirstRepo.git"
miheeka27/MyFirstRepo
https://github.com/Miheeka27/MyFirstRepo.git
miheeka27/student-management
https://github.com/Miheeka27/student-management.git
SURNAVSH112CODE/Old-India-SIF-Engine
https://github.com/SURNAVSH112CODE/Old-India-SIF-Engine.git
TanmayAhuja2006/hackathon-project
https://github.com/TanmayAhuja2006/hackathon-project.git
TrishKhatri/fresh-express-go
https://github.com/TrishKhatri/fresh-express-go.git
Rebuild every time a code change is pushed to this repository
Build type
Single build
Triggers a single build
Batch build
Triggers multiple builds as single execution
Comment approval
ALL_PULL_REQUESTS
Approver roles
CloudShell
Agent Toolkit for AWS
Feedback
Console Mobile App
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Choose pipeline settings Info

Step 2 of 7

Pipeline settings

Pipeline name

Enter the pipeline name. You cannot edit the pipeline name after it is created.

pwa-pipeline

No more than 100 characters

Execution mode Info

Choose the execution mode for your pipeline. This determines how the pipeline is run.

- ○ Superseded
- ◉ Queued
- ○ Parallel

Service role

- ◉ **New service role**
  Create a service role in your account
- ○ **Existing service role**
  Choose an existing service role from your account

Role name

AWSCodePipelineServiceRole-us-east-1-pwa-pipeline

Type your service role name

☑ Allow AWS CodePipeline to create a service role so it can be used with this new pipeline

▼ Advanced settings

Configure artifact store location, encryption settings, and pipeline variables for your pipeline.

Artifact store

- ○ **Default location**
  Create a default S3 bucket in your account.
- ◉ **Custom location**
  Choose an existing S3 location from your account in the same region and account as your pipeline

Bucket

🔍 store-build-artifact-vb ✕

Encryption key

- ◉ **Default AWS Managed Key**
  Use the AWS managed customer master key for CodePipeline in your account to encrypt the data in the artifact store.
- ○ **Customer Managed Key**
  To encrypt the data in the artifact store under an AWS KMS customer managed key, specify the key ID, key ARN, or alias ARN.

Variables

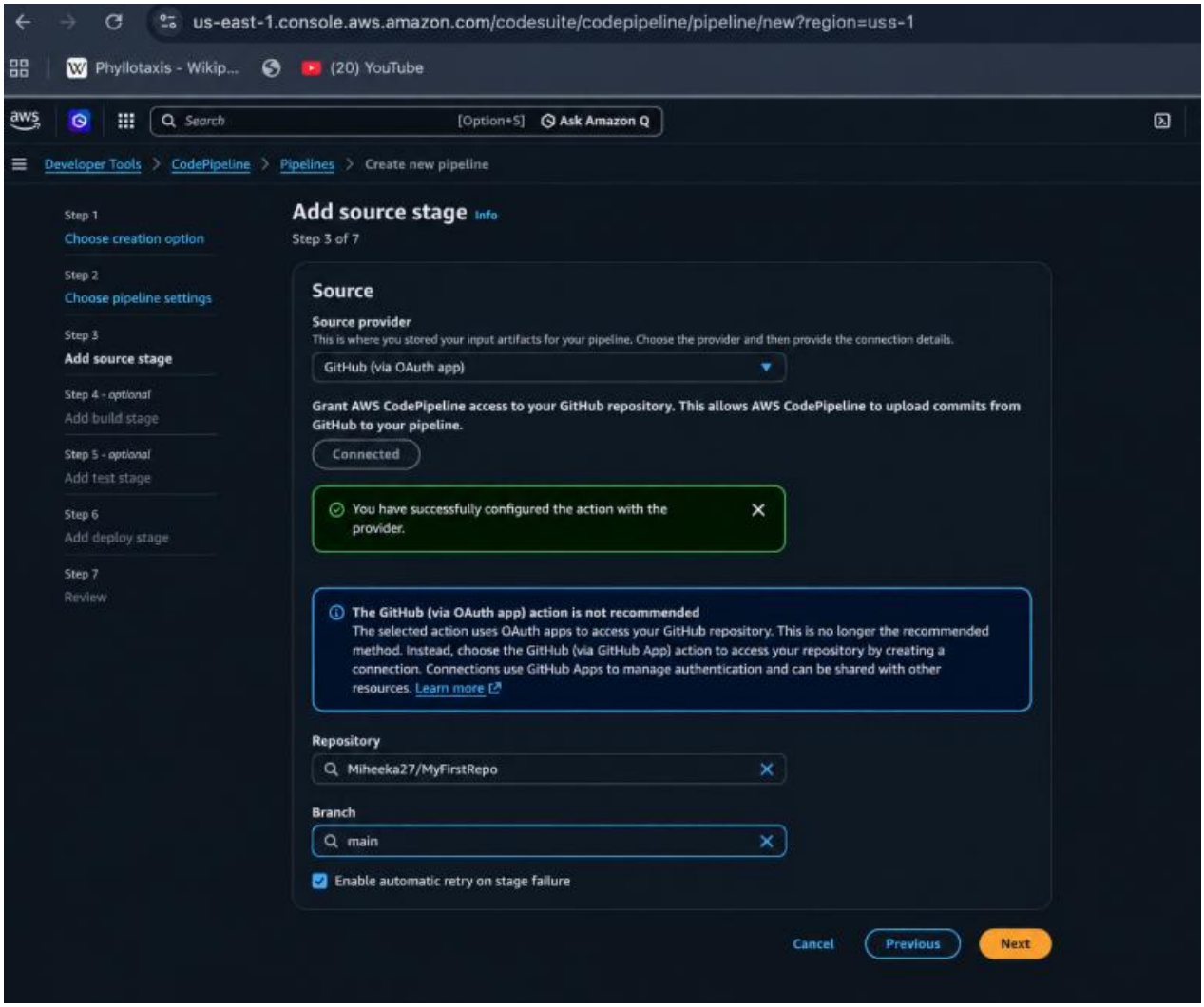
You can add variables at the pipeline level. You can choose to assign the value when you start the pipeline. Learn more

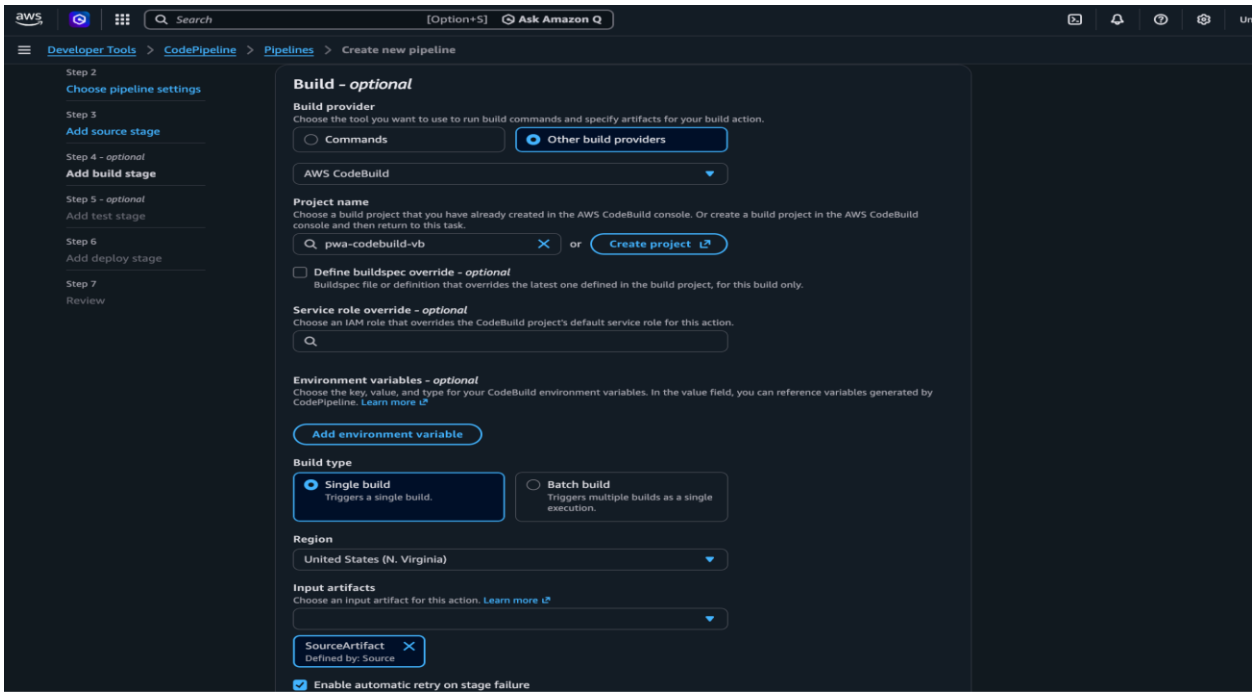
No variables defined at the pipeline level in this pipeline.

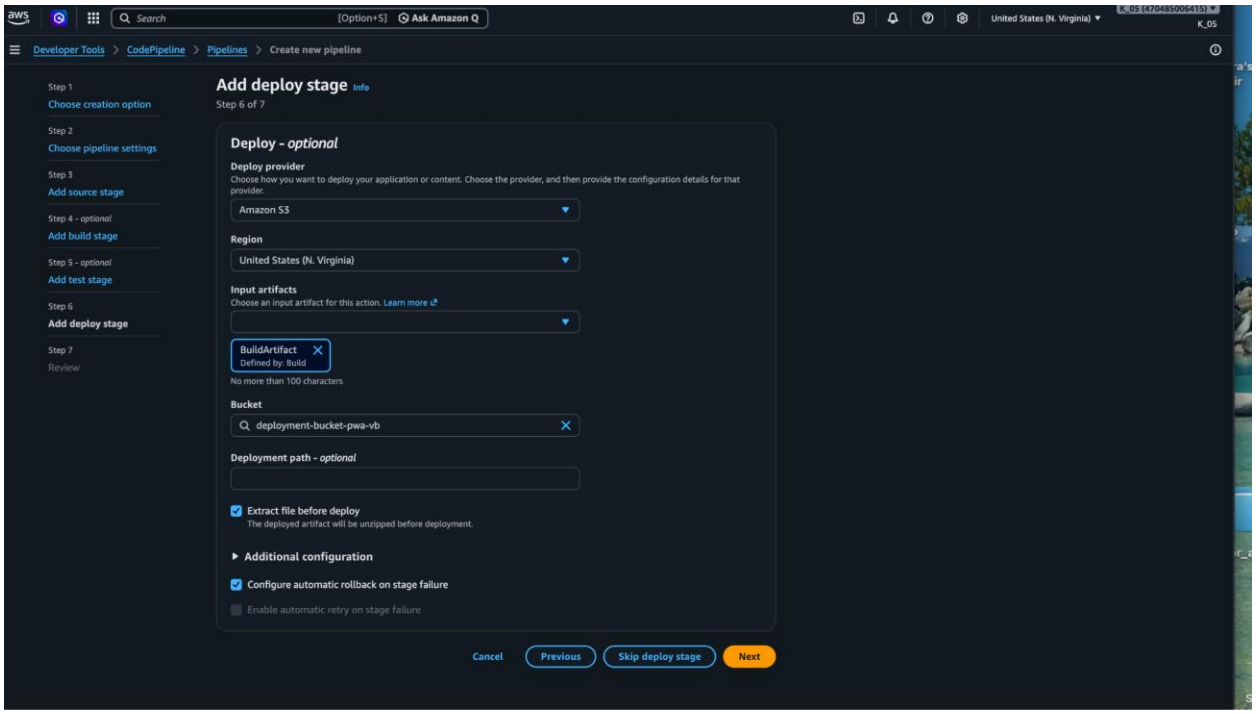
Add variable

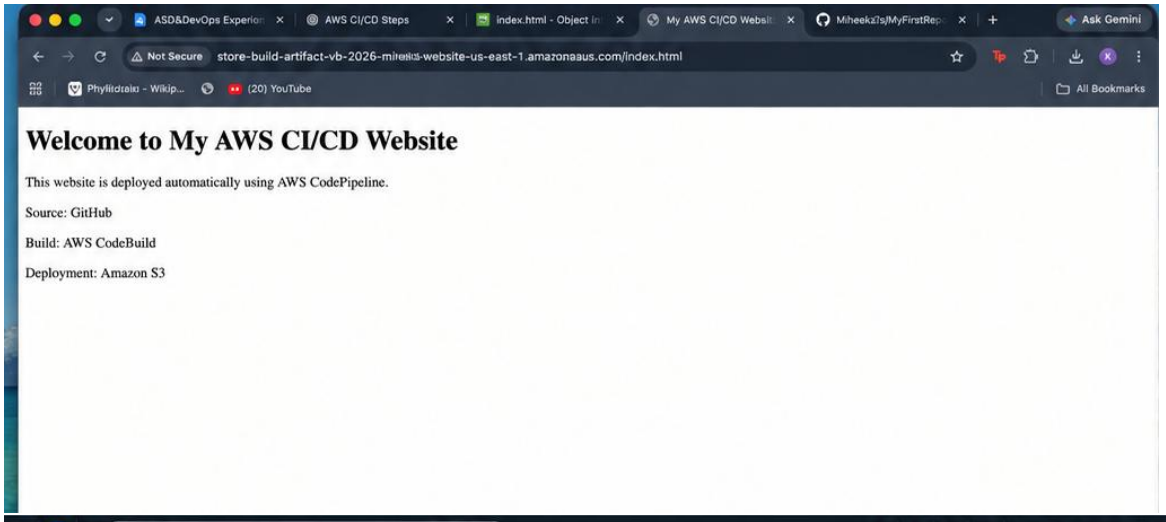
You can add up to 50 variables.

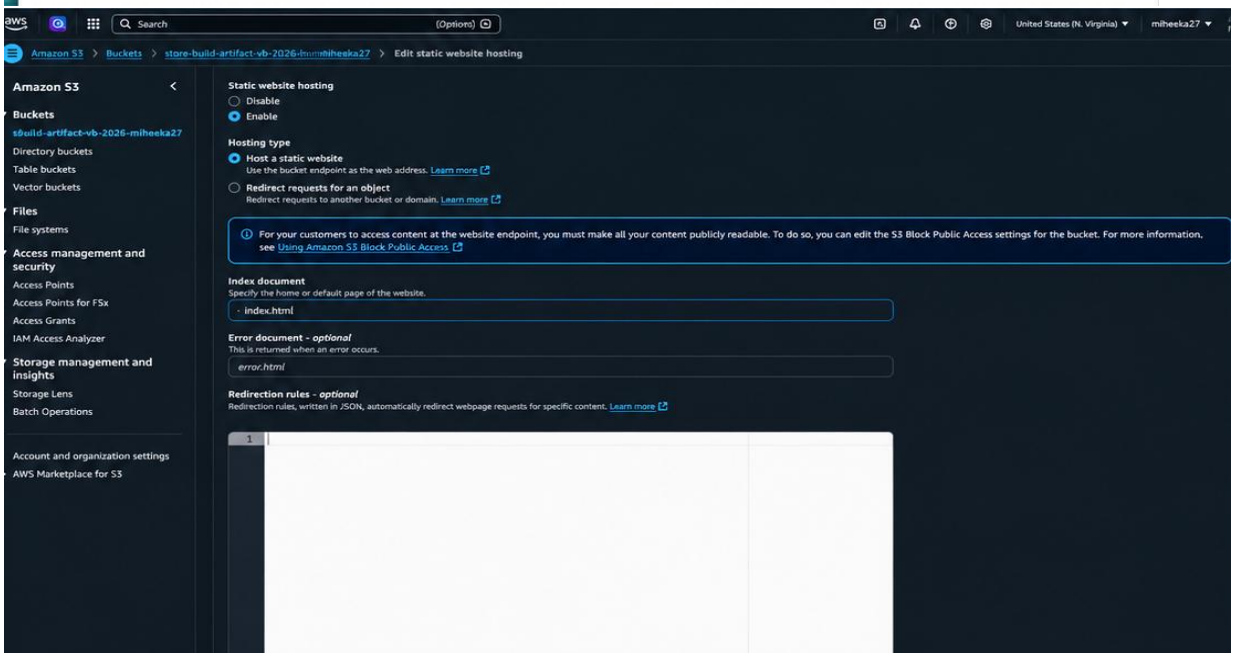
Cancel Previous Next

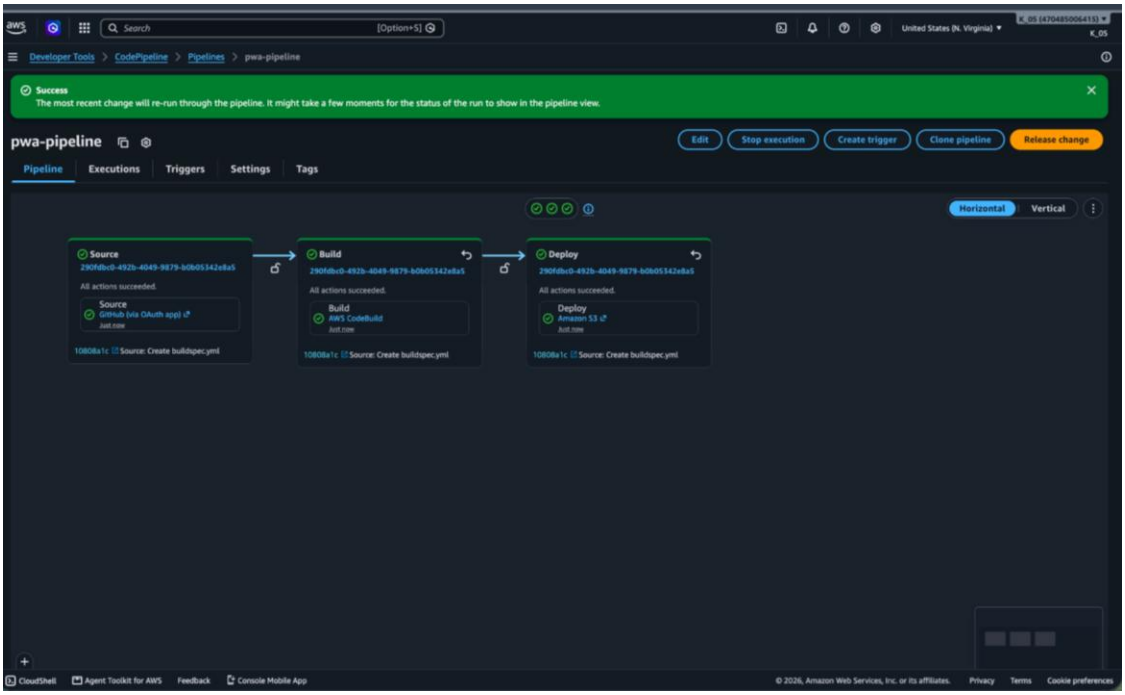

us-east-1.console.aws.amazon.com/codesuite/codepipeline/pipeline/new?region=us-1
Phyllotaxis - Wikip...
(20) YouTube
Search
[Option+S]
Ask Amazon Q
Developer Tools
CodePipeline
Pipelines
Create new pipeline
Step 1
Choose creation option
Step 2
Choose pipeline settings
Step 3
Add source stage
Step 4 - optional
Add build stage
Step 5 - optional
Add test stage
Step 6
Add deploy stage
Step 7
Review
Add source stage
Info
Step 3 of 7
Source
Source provider
This is where you stored your input artifacts for your pipeline. Choose the provider and then provide the connection details.
GitHub (via OAuth app)
Grant AWS CodePipeline access to your GitHub repository. This allows AWS CodePipeline to upload commits from GitHub to your pipeline.
Connected
You have successfully configured the action with the provider.
The GitHub (via OAuth app) action is not recommended
The selected action uses OAuth apps to access your GitHub repository. This is no longer the recommended method. Instead, choose the GitHub (via GitHub App) action to access your repository by creating a connection. Connections use GitHub Apps to manage authentication and can be shared with other resources. Learn more
Repository
Miheeka27/MyFirstRepo
Branch
main
Enable automatic retry on stage failure
Cancel
Previous
Next

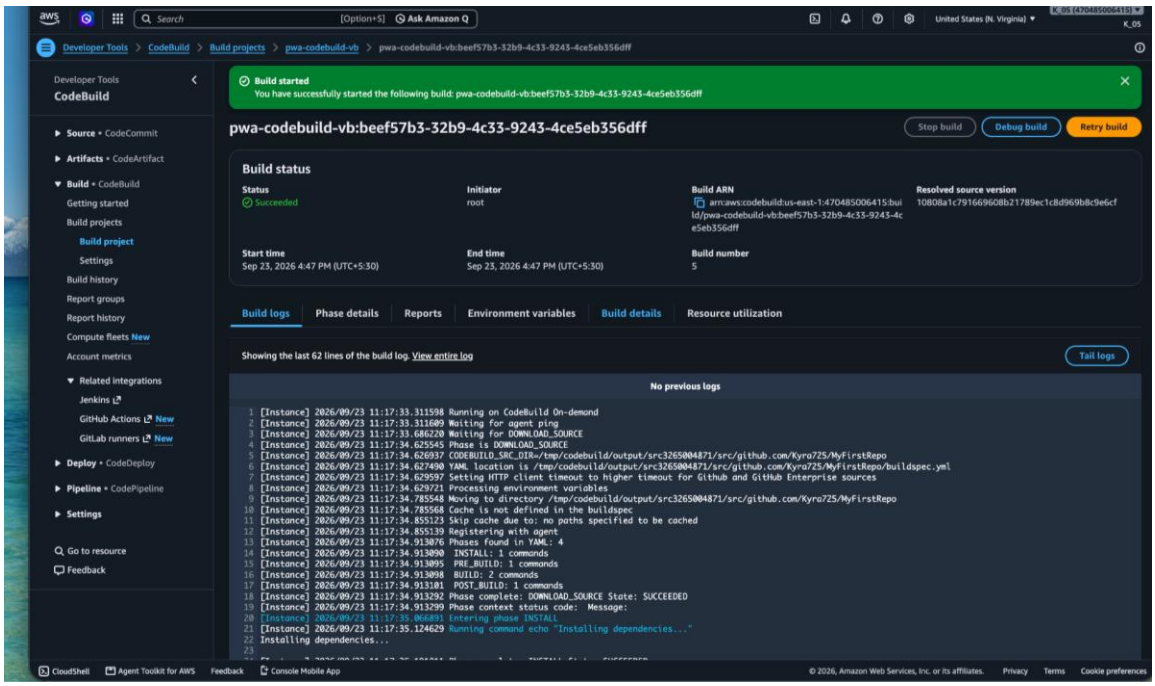

Search
[Option+S]
Ask Amazon Q
Developer Tools > CodePipeline > Pipelines > Create new pipeline
Step 2
Choose pipeline settings
Step 3
Add source stage
Step 4 - optional
Add build stage
Step 5 - optional
Add test stage
Step 6
Add deploy stage
Step 7
Review
Build - optional
Build provider
Choose the tool you want to use to run build commands and specify artifacts for your build action.
Commands
Other build providers
AWS CodeBuild
Project name
Choose a build project that you have already created in the AWS CodeBuild console. Or create a build project in the AWS CodeBuild console and then return to this task.
pwa-codebuild-vb
or
Create project
Define buildspec override - optional
Buildspec file or definition that overrides the latest one defined in the build project, for this build only.
Service role override - optional
Choose an IAM role that overrides the CodeBuild project's default service role for this action.
Environment variables - optional
Choose the key, value, and type for your CodeBuild environment variables. In the value field, you can reference variables generated by CodePipeline. Learn more
Add environment variable
Build type
Single build
Triggers a single build.
Batch build
Triggers multiple builds as a single execution.
Region
United States (N. Virginia)
Input artifacts
Choose an input artifact for this action. Learn more
SourceArtifact
Defined by: Source
Enable automatic retry on stage failure

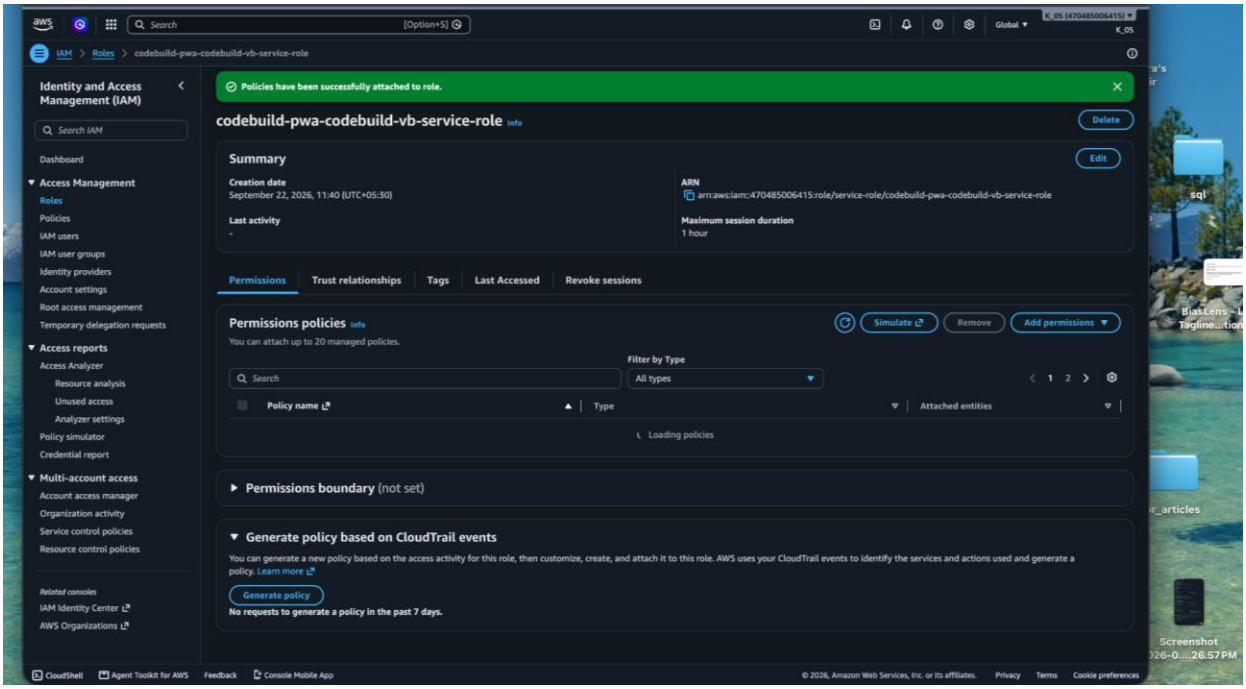

Search
[Option+S]
Ask Amazon Q
United States (N. Virginia)
K_05
Developer Tools > CodePipeline > Pipelines > Create new pipeline
Step 1
Choose creation option
Step 2
Choose pipeline settings
Step 3
Add source stage
Step 4 - optional
Add build stage
Step 5 - optional
Add test stage
Step 6
Add deploy stage
Step 7
Review
Add deploy stage Info
Step 6 of 7
Deploy - optional
Deploy provider
Choose how you want to deploy your application or content. Choose the provider, and then provide the configuration details for that provider.
Amazon S3
Region
United States (N. Virginia)
Input artifacts
Choose an input artifact for this action. Learn more
BuildArtifact
Defined by: Build
No more than 100 characters
Bucket
deployment-bucket-pwa-vb
Deployment path - optional
Extract file before deploy
The deployed artifact will be unzipped before deployment.
Additional configuration
Configure automatic rollback on stage failure
Enable automatic retry on stage failure
Cancel
Previous
Skip deploy stage
Next

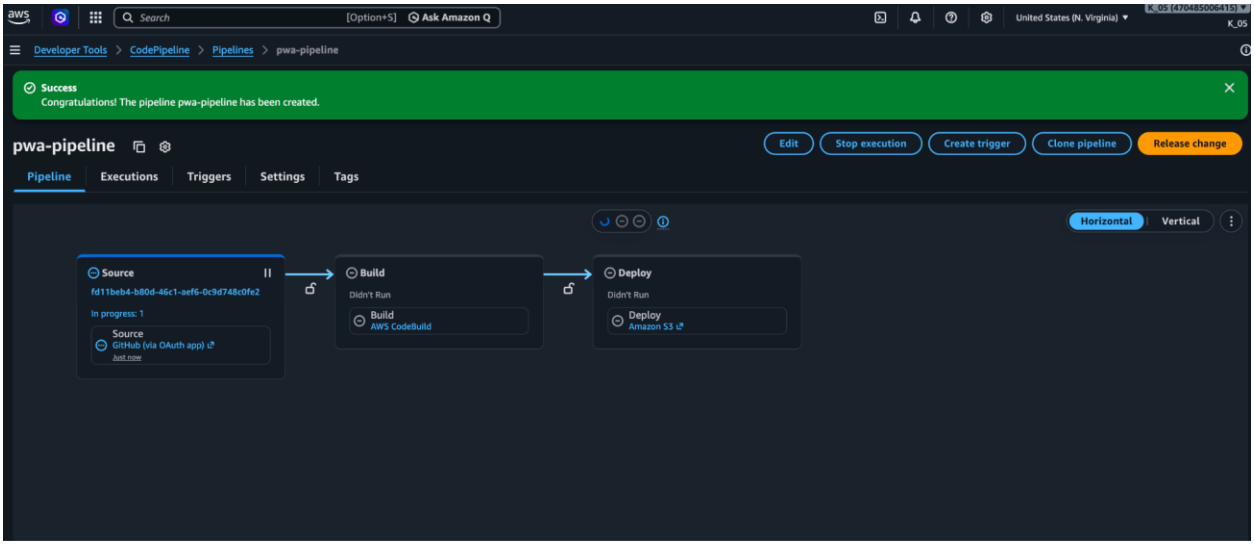

ASD&DevOps Experien
AWS CI/CD Steps
index.html - Object in
My AWS CI/CD Websit
MiheekaTs/MyFirstRep
Ask Gemini
Not Secure
store-build-artifact-vb-2026-miheeka3-website-us-east-1.amazonaaus.com/index.html
Phyllodraka - Wikip...
(20) YouTube
All Bookmarks
Welcome to My AWS CI/CD Website
This website is deployed automatically using AWS CodePipeline.
Source: GitHub
Build: AWS CodeBuild
Deployment: Amazon S3


Search
(Options)
United States (N. Virginia)
miheeka27
Amazon S3 > Buckets > store-build-artifact-vb-2026-lnmmiheeka27 > Edit static website hosting
Amazon S3
Buckets
sBuild-artifact-vb-2026-miheeka27
Directory buckets
Table buckets
Vector buckets
Files
File systems
Access management and security
Access Points
Access Points for FSx
Access Grants
IAM Access Analyzer
Storage management and insights
Storage Lens
Batch Operations
Account and organization settings
AWS Marketplace for S3
Static website hosting
Disable
Enable
Hosting type
Host a static website
Use the bucket endpoint as the web address. Learn more
Redirect requests for an object
Redirect requests to another bucket or domain. Learn more
For your customers to access content at the website endpoint, you must make all your content publicly readable. To do so, you can edit the S3 Block Public Access settings for the bucket. For more information, see Using Amazon S3 Block Public Access
Index document
Specify the home or default page of the website.
index.html
Error document - optional
This is returned when an error occurs.
error.html
Redirection rules - optional
Redirection rules, written in JSON, automatically redirect webpage requests for specific content. Learn more
1


Developer Tools > CodePipeline > Pipelines > pwa-pipeline
Success
The most recent change will re-run through the pipeline. It might take a few moments for the status of the run to show in the pipeline view.
pwa-pipeline
Edit
Stop execution
Create trigger
Clone pipeline
Release change
Pipeline
Executions
Triggers
Settings
Tags
Horizontal
Vertical
Source
Build
Deploy
All actions succeeded.


Developer Tools > CodeBuild > Build projects > pwa-codebuild-vb > pwa-codebuild-vb:beef57b3-32b9-4c33-9243-4ce5eb356dff
Build started
You have successfully started the following build: pwa-codebuild-vb:beef57b3-32b9-4c33-9243-4ce5eb356dff
pwa-codebuild-vb:beef57b3-32b9-4c33-9243-4ce5eb356dff
Stop build
Debug build
Retry build
Build status
Status
Succeeded
Initiator
root
Build ARN
Resolved source version
Start time
Sep 23, 2026 4:47 PM (UTC+5:30)
End time
Sep 23, 2026 4:47 PM (UTC+5:30)
Build number
5
Build logs
Phase details
Reports
Environment variables
Build details
Resource utilization
Showing the last 62 lines of the build log. View entire log
Tail logs
No previous logs
[Instance] 2026/09/23 11:17:33.311598 Running on CodeBuild On-demand
[Instance] 2026/09/23 11:17:33.311609 Waiting for agent ping
[Instance] 2026/09/23 11:17:33.686220 Waiting for DOWNLOAD_SOURCE
[Instance] 2026/09/23 11:17:34.625545 Phase is DOWNLOAD_SOURCE
[Instance] 2026/09/23 11:17:34.626937 CODEBUILD_SRC_DIR=/tmp/codebuild/output/src3265004871/src/github.com/Kyra725/MyFirstRepo
[Instance] 2026/09/23 11:17:34.627490 YAML location is /tmp/codebuild/output/src3265004871/src/github.com/Kyra725/MyFirstRepo/buildspec.yml
[Instance] 2026/09/23 11:17:34.629597 Setting HTTP client timeout to higher timeout for Github and GitHub Enterprise sources
[Instance] 2026/09/23 11:17:34.629721 Processing environment variables
[Instance] 2026/09/23 11:17:34.785548 Moving to directory /tmp/codebuild/output/src3265004871/src/github.com/Kyra725/MyFirstRepo
[Instance] 2026/09/23 11:17:34.785568 Cache is not defined in the buildspec
[Instance] 2026/09/23 11:17:34.855123 Skip cache due to: no paths specified to be cached
[Instance] 2026/09/23 11:17:34.855139 Registering with agent
[Instance] 2026/09/23 11:17:34.913076 Phases found in YAML: 4
[Instance] 2026/09/23 11:17:34.913090 INSTALL: 1 commands
[Instance] 2026/09/23 11:17:34.913095 PRE_BUILD: 1 commands
[Instance] 2026/09/23 11:17:34.913098 BUILD: 2 commands
[Instance] 2026/09/23 11:17:34.913101 POST_BUILD: 1 commands
[Instance] 2026/09/23 11:17:34.913292 Phase complete: DOWNLOAD_SOURCE State: SUCCEEDED
[Instance] 2026/09/23 11:17:34.913299 Phase context status code: Message:
[Instance] 2026/09/23 11:17:35.066891 Entering phase INSTALL
[Instance] 2026/09/23 11:17:35.124629 Running command echo "Installing dependencies..."
Installing dependencies...





Conclusion:

LO mapped successfully.